The AI dev toolchain

Task 1: Explore the Tools

1. IDE (Integrated Development Environment)

An IDE is a program that programmers use to write code. It is not just a text editor; it provides a complete environment that includes tools like running programs, debugging (fixing errors), a built-in terminal, and project management. It also supports extensions to customize it based on the developer's needs, which makes programming easier and more efficient.

2. CLI (Command Line Interface)

A CLI is a text-based interface. It relies on entering text commands instead of using a Graphical User Interface (GUI). Through it, the user can run commands, start programs, and interact with the operating system via the Shell. The Shell interprets these commands and sends them to the operating system to be executed. The CLI is fast, lightweight, and uses fewer resources than a GUI. It is also very useful for automating tasks and writing scripts. However, it requires knowing the specific commands and how to use them, so it is more suitable for developers, system administrators, and advanced users rather than regular users.

3. API (Application Programming Interface)

An API is an interface for communication between different applications and systems. The user sends a request through the application, the API carries it to the server, and then the response comes back from the server to the application to show it to the user. In AI applications, the API works in the exact same way. The application sends the prompt (the request) to the model's server via the API. The model processes this request, and then the server sends back the response containing the text, image, or the required result to the application. Using an API helps separate the Frontend from the Backend, making development, maintenance, and updates much easier. It also allows reusing the same services across multiple different applications without needing to repeat code.

Task 2: Draw the Workflow

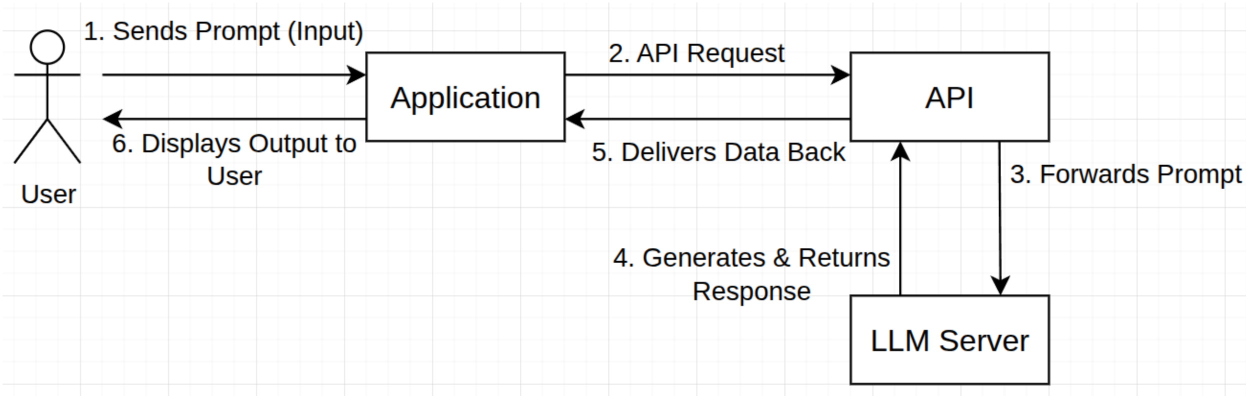

Figure 1: AI Development Workflow and Data Flow Architecture

Workflow Steps:

1. Sends Prompt (Input): The User interacts with the Application and types an instruction or question (the Prompt).
2. API Request: The Application takes the user's prompt, packages it, and sends it as a request through the API.
3. Forwards Prompt: The API acts as a bridge and carries the prompt over the internet to the cloud where the LLM Server lives.
4. Generates & Returns Response: The LLM Server processes the prompt, creates the answer, and sends it back to the API.
5. Delivers Data Back: The API receives the answer from the server and delivers it back to the Application.
6. Displays Output to User: The Application receives the final data and displays it nicely on the screen for the User to see.

Practical Examples:

Example 1: ChatGPT (Web Interface)

1. User: A person who wants to write an essay.
2. Prompt: "Write a 300-word essay about AI."
3. Application: The official ChatGPT website that the user sees and types into.
4. API: The hidden internal connection that OpenAI uses to pass the text from the website to the model.
5. LLM Server: The powerful remote servers running the GPT model that actually writes the essay.

Example 2: A Python Script that Calls an API

1. User: A developer running a script.
2. Prompt: A text string saved inside a variable in the Python code.
3. Application: The Python script itself running inside an IDE or launched via the CLI (Terminal).
4. API: A code library (like the openai Python package) that sends the prompt directly from the script to the server.
5. LLM Server: The cloud model that processes the script's request and sends back the response to be printed on the Terminal.